Lab 4: VibeCoding - Tug of War Repair Lab

Pair-Programming & Prompt Engineering Chat History

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Assigned Repo: github.com/SETAPESU26/56_tug_of_war

Target Repo: github.com/Suryabhat48/Software_Engineering

1. Deliverables Checklist

[X] a. Videos (before & after): gameplay_before.mp4 (deadlock) & gameplay_after.mp4 (all features)

[X] b. Updated Code: Lab-4/main.py, Lab-4/game/game_engine.py, player.py, rope.py

[X] c. Chat History Document: chat_history.pdf & chat_history.docx

[X] Independent Git Commits: Separate commit history for Task 1, Task 2, Task 3, Task 4

Task 1: Fix Alternating Input Deadlock Bug

Prompt to AI:

Analyze input handling in game/game_engine.py. Rapidly mashing between 'A' and 'D' freezes input completely due to lock state. Redesign input synchronization so overlapping keystrokes consistently register pulls.

AI Pair Programmer Solution & Implementation:

Root Cause: KEYUP dependency locked input when rapid presses overlapped.

Fix: Decoupled pull from KEYUP; enforced alternation on KEYDOWN while ignoring single-key repeats. Result: Zero dropped inputs during rapid mashing.

Task 2: Dynamic AI Panic Surges

Prompt to AI:

Implement dynamic difficulty: as the center marker gets pulled closer to the player's winning threshold, have the computer enter a panic surge with faster reaction cadence and aggressive pulling strength.

AI Pair Programmer Solution & Implementation:

Fix: Computed dynamic panic_level based on proximity to left_win_x. Cadence accelerates from 180ms down to 65ms, pull strength increases by up to +120%, and visual HUD banner indicates surge mode.

Task 3: Dynamic Rope Tension, Hum & Puller Leaning

Prompt to AI:

Give the rope a vibrating high-tension hum or sag depending on struggle intensity, and animate both characters with backward leaning postures based on pulling momentum.

AI Pair Programmer Solution & Implementation:

Fix: Puller character dynamically leans backward up to 32px based on momentum impulse with damped physics. Rope renders parabolic catenary sag under low tension and rapid wave oscillation (vibrating hum) under struggle.

Task 4: Match Timer and Sudden Death Mode

Prompt to AI:

Add a live match timer at the top of the screen. If no side has won within 45 seconds, trigger Sudden Death mode doubling pulling power across all actions.

AI Pair Programmer Solution & Implementation:

Fix: Real-time match timer HUD implemented. At 45.0s, is_sudden_death triggers 2x pull multiplier across all player and AI actions with flashing gold/red banner alert.